

# Stellar\_ImportIOCs

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Scripts (PowerShell, Python, Bash) that import file-hash IOCs from a CSV file into **TXOne StellarOne** as User-Defined Suspicious Objects (UDSOs) via the StellarOne REST API.

In TXOne StellarOne, the global policy for each product type contains a list of User-Defined Suspicious Objects — file hashes that the agent should watch for on managed endpoints. When a running file matches a hash in this list, StellarOne logs a detection event and can block execution. These scripts automate loading a fresh IOC list into StellarOne in seconds, without touching the management console.

## Step-by-Step: What happens when you run a script

| Step | What the script does                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------|
| 1    | Reads the StellarOne server address and API key from <code>StellarOne.conf</code> .                           |
| 2    | Parses the IOC CSV file, skipping blank lines, comment lines, and the optional header row.                    |
| 3    | Auto-detects the hash type for each row from the hash length (SHA-1 = 40 hex chars, SHA-256 = 64 hex chars).  |
| 4    | Skips MD5 hashes (32 hex chars) with a warning — StellarOne's UDSO API does not support MD5.                  |
| 5    | For each product type (StellarProtect and StellarProtect Legacy Mode), <b>GETs</b> the current global policy. |
| 6    | Merges the new hashes into the existing UDSO list — existing hashes are preserved and duplicates are skipped. |
| 7    | <b>PUTs</b> the updated policy back to StellarOne.                                                            |
| 8    | Prints a summary: how many hashes were added, already present, skipped, or failed.                            |

## Prerequisites

All scripts require:

- **Network access** to the StellarOne management server
- `StellarOne.conf` in the same folder as the script (see Setup below)
- `IOCs.csv` — the IOC list to import (see CSV Format below)

Additional prerequisites per script:

| Script                 | Runtime required                                                   |
|------------------------|--------------------------------------------------------------------|
| Stellar_ImportIOCs.ps1 | Windows PowerShell 5.1 (built into Windows 10 / Server 2016+)      |
| Stellar_ImportIOCs.py  | Python 3.6 or later — standard library only, no pip install needed |
| Stellar_ImportIOCs.sh  | bash 4.0+, curl, and python3 (used as JSON processor)              |

**Tested on:** PowerShell 5.1 / Windows 10 · Python 3.12 / Windows MINGW64 · bash 5.2 / Windows MINGW64 and bash 5.1 / Ubuntu 22.04 — all against TXOne StellarOne 3.3.1392.

## Setup

1. Copy stellarOne\_example.conf to StellarOne.conf
2. Fill in your StellarOne server URL and API key:

```
StellarOneURL="https://YOUR_STELLARONE_IP_OR_HOSTNAME"
ApiKey="YOUR_STELLARONE_API_KEY_HERE"
```

3. Create your IOCs.csv file (or copy from IOCs\_example.csv and replace the hashes with real IOCs).

**Security note:** StellarOne.conf contains sensitive credentials. Restrict access to this file to authorised administrators only. Never commit it to version control.

## CSV Format

```
# Lines starting with '#' are treated as comments and ignored.
# Blank lines are also ignored.
# An optional header row (first cell = "hash") is automatically skipped.
# The action column is optional and is accepted for documentation only.
# NOTE: MD5 hashes (32 hex chars) are NOT supported and will be skipped.
#
hash,description,action
935200fdb9410116dea80d66b1349d3fca9f4d8b,Ransomware dropper - ticket #1234 (SHA-1),BLOCK
e46b6e54f91e82d8ec5853764ed8d226912de90a,Dropper variant x64 (SHA-1),BLOCK
906895fbd6244c6d1cfc42a72e0e30100a237e8da61e374d8319a53cb29f81a0,Suspicious payload - log only (SHA-256),LOG
```

| Column      | Required | Description                                                                                                                                                                                          |
|-------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hash        | Yes      | SHA-1 (40 hex chars) or SHA-256 (64 hex chars). MD5 (32 hex chars) is <b>not</b> supported by StellarOne's UDSO API and will be skipped.                                                             |
| description | Yes      | Free-text note stored alongside the hash in StellarOne                                                                                                                                               |
| action      | No       | BLOCK or LOG. <b>Accepted for documentation only — not sent to the API.</b> StellarOne does not support per-hash actions; the action is a policy-level setting configured in the StellarOne console. |

Rows with an unrecognised hash length or non-hex characters are skipped with a warning — the rest of the file continues to import.

## How to run

```
# PowerShell -- use the default IOCs.csv in the script folder
.\Stellar_ImportIOCs.ps1

# PowerShell -- pass an explicit path
.\Stellar_ImportIOCs.ps1 -CsvPath "C:\IOCs\campaign_xyz.csv"

# Python -- use the default IOCs.csv in the script folder
python Stellar_ImportIOCs.py

# Python -- pass an explicit path
python Stellar_ImportIOCs.py /path/to/my_iocs.csv

# Bash -- use the default IOCs.csv in the script folder
bash Stellar_ImportIOCs.sh

# Bash -- pass an explicit path
bash Stellar_ImportIOCs.sh /path/to/my_iocs.csv
```

## Parameters

| Parameter | Required | Description                                                          |
|-----------|----------|----------------------------------------------------------------------|
| CSV path  | No       | Path to the IOC CSV file. Defaults to IOCs.csv in the script folder. |

## StellarOne API calls used

### 1 — Read the current global policy

GET /api/v1/policy/global/{product\_code}

| {product_code} | Product                    |
|----------------|----------------------------|
| PRODUCT_SP     | StellarProtect             |
| PRODUCT_SPLM   | StellarProtect Legacy Mode |

Returns the full global policy for that product type, including the existing userDefinedSuspiciousObjects list.

## 2 — Write the updated global policy

PUT /api/v1/policy/global

Only the userDefinedSuspiciousObjects block is sent in the request body. All other policy settings are left untouched (replace-on-presence semantics).

### Request body structure:

```
{
  "policy": {
    "spPolicy": {
      "userDefinedSuspiciousObjects": {
        "list": [
          {
            "value": "<hash in lowercase>",
            "notes": "<description from CSV>",
            "type": "TYPE_SHA1"
          },
          {
            "value": "<hash in lowercase>",
            "notes": "<description from CSV>",
            "type": "TYPE_SHA256"
          }
        ]
      }
    }
  }
}
```

| Field | Value                    | Notes                                  |
|-------|--------------------------|----------------------------------------|
| value | Hash value               | Normalised to lowercase before sending |
| notes | Description from the CSV | Stored as a note in StellarOne         |
| type  | TYPE_SHA1 or TYPE_SHA256 | Auto-detected from hash length         |

**Note:** Use splmPolicy instead of spPolicy when updating the StellarProtect Legacy Mode policy.

## Known Limitations

- **MD5 hashes are not supported.** The StellarOne UDSO API only accepts SHA-1 and SHA-256. MD5 rows in the CSV are skipped with a warning.

- **No per-hash BLOCK/LOG action.** StellarOne does not support configuring the action per individual hash. The action column in the CSV is accepted for documentation purposes but is not sent to the API. Configure the response action in the StellarOne console at the policy level.

## Troubleshooting

| Problem                                | Likely cause                                         | Solution                                                        |
|----------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|
| IOC file not found                     | CSV file is missing or path is wrong                 | Check the file exists and re-run with an explicit path          |
| not a recognised SHA-1 or SHA-256 hash | Hash has wrong length or contains non-hex characters | Fix the row in the CSV                                          |
| MD5 hash not supported                 | 32-character hex hash detected                       | Replace MD5 hashes with SHA-1 or SHA-256 equivalents            |
| API call failed - HTTP 401             | API key is wrong or expired                          | Regenerate the key in StellarOne and update StellarOne.conf     |
| API call failed - HTTP 403             | API key lacks write permission                       | Ask your StellarOne administrator to grant the key write access |
| Could not connect to ...               | Wrong server address or firewall blocking port 443   | Verify the URL in StellarOne.conf and network connectivity      |
| No global policy found for ...         | Product type has no global policy configured yet     | Configure a global policy in the StellarOne console first       |

## Files

```
StellarAPI/
  Stellar_ImportIOCs.ps1 <-- PowerShell 5.1 script
  Stellar_ImportIOCs.py  <-- Python 3 script
  Stellar_ImportIOCs.sh  <-- Bash script
  Stellar_ImportIOCs.md  <-- This documentation
  Stellar_ImportIOCs.pdf <-- PDF version of this documentation
  IOCs.csv               <-- Your IOC list (keep this file private!)
  IOCs_example.csv       <-- Template -- copy to IOCs.csv and fill in real hashes
  StellarOne.conf         <-- Server URL + API key (keep this file private!)
  stellarOne_example.conf <-- Template -- copy to StellarOne.conf and fill in values
```